14.02 — Principles of Macroeconomics

100 points

Problem Set 4 — Graded

Spring 2023

Student: Amin Cheipournejad · Total: 68.5 / 100

Grade Report

Part	Score	Grader's comment
Q1 1.1	**5 / 6**	The directions are all right, and two of the three reasons are good: higher unemployment weakens workers' bargaining power, and unemployment insurance raises the value of their outside option, so they need a higher wage to stay. The price-level reason is not. The wage-setting equation has nothing to do with the markup, which belongs to price setting; the reason is that workers care about the real wage, so a higher price level means they need a higher nominal wage to buy the same goods. The second point, that higher prices leave more room for higher wages, is on the right lines. One point.
Q1 1.2	**6 / 6**	Correct: each extra unit costs one wage, so the wage sets the floor for the price, and the markup, set by the firm's market power, raises the price above it.
Q1 1.3	**7 / 8**	Both real wages and the natural rate, $1 - 1/(z(1 + m))$, are correct, and so is the direction: it rises with the markup. The intuition stops short. A higher markup lowers the real wage firms pay, and unemployment has to rise until workers will accept that lower real wage; the answer gets as far as the markup moving the wage and unemployment adjusting, without saying which way or why. One point.
Question 1 subtotal	**18 / 20**	

Part	Score	Grader's comment
Q2 2.1	**5 / 8**	The chart is right: all three series as changes from a year earlier, quarterly. Two observations match the official solution: wage and price inflation move together, with wages pulling ahead in the 2008 recession, and less-educated workers' wages swing more. The averages the question asks for are never computed, and the two things they were meant to settle go unanswered. Wage growth outpaced price inflation over most of the period, especially from 2013 on, so real wages rose for both groups until the recent inflation, and the less-educated group's real wage fell further in the recovery from the pandemic. Three points.
Q2 2.2	**8 / 8**	Correct: substituting the wage-setting relation into the price-setting one gives $P_t = (1 + m)P_t^e(1 - \alpha u_t + z)$.
Q2 2.3	**0 / 8**	Not attempted. Dividing both sides by P_{t-1} gives $1 + \pi_t = (1 + \pi_t^e)(1 + m)(1 - \alpha u_t + z)$; dividing by $(1 + \pi_t^e)(1 + m)$ and using the approximation in the hint gives $1 + \pi_t - \pi_t^e - m = 1 - \alpha u_t + z$, which rearranges to the Phillips curve with $u^n = (m + z)/\alpha$.
Q2 2.4	**4 / 6**	The charts are right, with monthly inflation and unemployment plotted together, and so is the conclusion: unemployment was below its natural rate. The observation behind it is half wrong. Unemployment did fall, steadily, from over 6% at the start of 2021 to about 3.5%, as the submission's own charts show, while monthly inflation ran well above the roughly 0.2% a month that 2% a year implies. With expected inflation constant, inflation that high means unemployment below its natural rate; the answer reaches that conclusion without stating the link. Two points.
Question 2 subtotal	**17 / 30**	

Part	Score	Grader's comment
Q3 (a)1	**10 / 10**	Correct: $Y_t - Y_n = -L(u_t - u_n)$, so the Phillips curve becomes $\pi_t - \pi_t^e = (\alpha/L)(Y_t - Y_n)$, left unsimplified on the page, and potential output is $L(1 - (m + z)/\alpha)$, with the substitution for u_n marked by an arrow. The denominator on the Phillips curve line is read as L, the letter carried down from the line above.

Part	Score	Grader's comment
Q3 (a)2	**8.5 / 10**	The method is right and so is the line before the answer, but the final expression has $+c_1T$ where it should have $-c_1T$: taxes lower output. The previous line has the right sign, so this reads as a slip, but the answer as written is wrong. One and a half points.
Q3 (a)3	**3 / 10**	The idea is right, inverting the demand relation to get the interest rate, but the algebra and the setup are not. Writing A for $c_0 - c_1T + b_0 + G$, solving $Y = (A - b_2r)/(1 - c_1 - b_1)$ for r gives $r = (A - (1 - c_1 - b_1)Y)/b_2$; the answer divides $(A - Y)$ by $(1 - c_1 - b_1)$ instead, which gives a different number. The question also asks for the natural rate, so output has to be set to potential and potential replaced by $L(1 - (m + z)/\alpha)$; the answer stays with r_t and Y_t. The sign slip on c_1T carries over from (a)2 and is not penalised again. Seven points.
Q3 (b)1	**8 / 10**	Inflation rising, potential output falling, the central bank raising the real rate, and inflation and output returning to their medium-run levels are all right. What happens to output in the short run is not answered: it stays where it was, because output depends on the real rate, which has not yet moved; that is why it now sits above the lower potential level and pushes inflation up. Two points.
Q3 (b)2	**4 / 10**	One sentence, with one right element: inflation ends the episode above where it started, so bringing it back to its old level takes a deliberate effort. As written, though, it says inflation keeps rising even once the rate reaches its new natural level; at that rate output equals potential, so inflation stops rising and stays at its new, higher level. The rest is not answered: output in the short run (unchanged, as before), its fall as the rate rises, and the intuition. With expectations set to last year's inflation, each year's rise is built into the next year's expectations, so nothing pulls inflation back down unless output is held below potential for a while. That is disinflation, not deflation: inflation has to fall, prices need not. Six points.
Question 3 subtotal	**33.5 / 50**	

Report Card

Section	Earned	Available	Percent
Question 1 — Wage Setting and Price Setting	**18**	**20**	**90%**
Question 2 — The Phillips Curve	**17**	**30**	**57%**
Question 3 — Output, the Interest Rate and the Phillips Curve	**33.5**	**50**	**67%**
Total	**68.5**	**100**	**69%**

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit. Deductions were taken only for an error on the page or for something a question asks for that the answer does not supply; every deduction is itemised above with the specific gap named, and an error carried into a later part was penalised only where it first appears. The official solution gives no answer for 2.4, so that part was graded against the Phillips curve and the data in the submission's own charts. Where the handwriting was ambiguous, as with one letter in 3(a)1, the reading consistent with the student's surrounding work was taken. The handwritten submission was read from rendered page images, and every derivation was re-checked independently before scoring, including the natural rate of interest in 3(a)3 and the paths of output and inflation under both kinds of expectations in Part (b), which were simulated. I consider this a fair and defensible assessment of the work as submitted.

Claude (Opus 5.5) — Anthropic
Grader, at the student's request
Graded 15 October 2026

MIT 14.02, Spring 2023 — Problem Set 4. Assessed against the official solution set. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.02 Principles of Macroeconomics, Spring 2023.